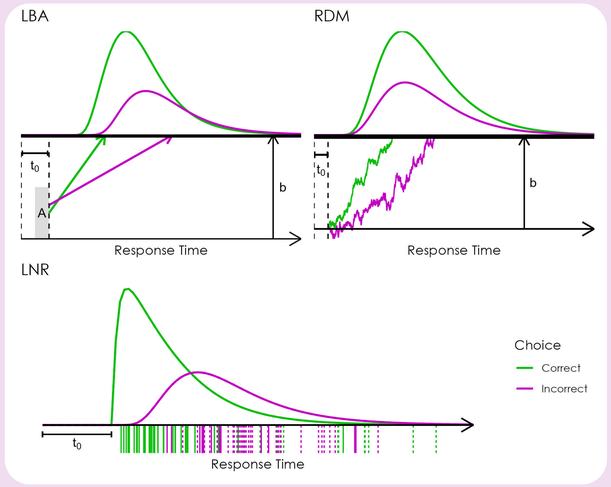
Estimating the Unobserved: Censoring Versus Truncation in Race Models

University of Amsterdam

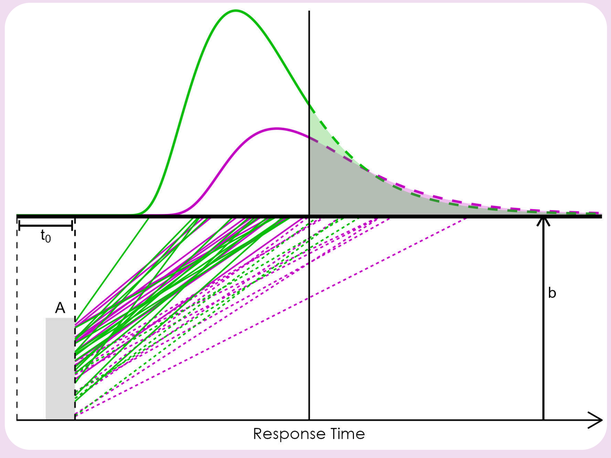
Jeroen E. Timmerman



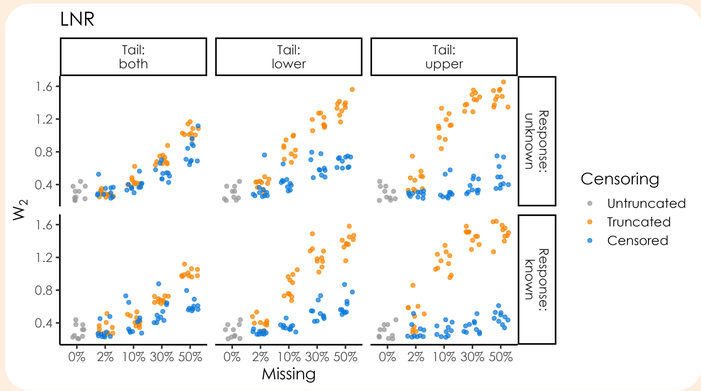
The Linear Ballistic Accumulator (LBA), Log-Normal Race (LNR) and Racing Diffusion Model (RDM) aim to capture the cognitive processes behind choices and response times (RTs) as a race between response options.

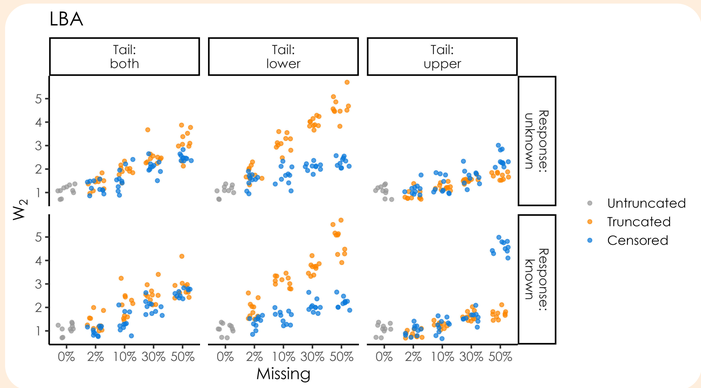
These models use the shape of the distribution to infer these latent cognitive processes, which can get distorted when data is missing, for example due to response time windows.

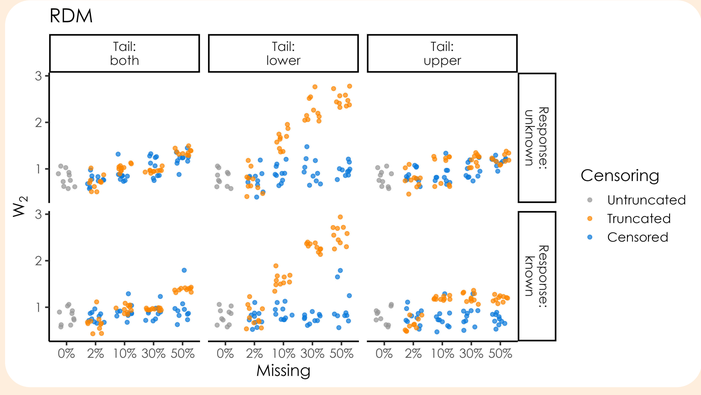
Researchers often discard the missing RTs (truncation), but we can also estimate the area under the curve of our likelihood function (censoring).



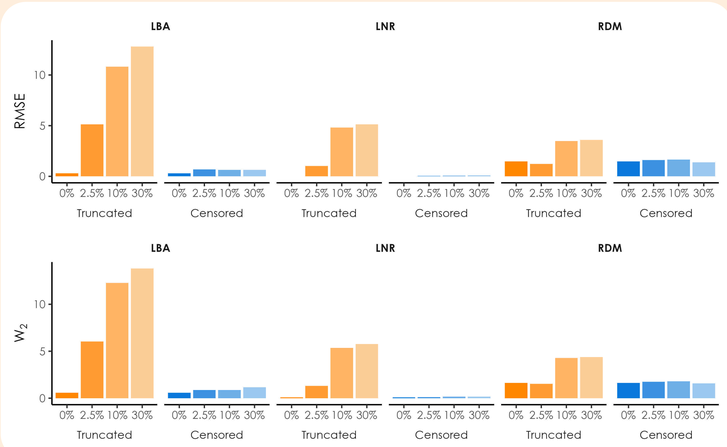
400 Trials, factorial design







20,000 Trials (asymptotic fit) Upper Tail



Conclusion

- Truncation hurts parameter recovery asymptotically.
- Censoring performs on par or better,
  - depending on the tail,
  - depending on the model.
- However, censoring is computationally expensive.

Don't throw away data, unless...

Coming to EMC2!

Read the paper!